Ericsson PE Node BAS-PE-10: Hardware/Environmental Fault Detection and Resolution at Basavanagudi POP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_7002

Description

This Standard Operating Procedure (SOP) outlines the steps for detecting and resolving hardware/environmental faults in the BAS-PE-10 node, located at the Basavanagudi POP, which is a critical PE (Packet Edge) node in the network. The node operates on Ericsson equipment, and the failure domain under investigation is related to Hardware/Environmental Fault, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss. The alarm code ALARM_7002 is triggered when a major hardware or environmental fault is detected, indicating a potential disruption to the network. The underlying protocol or hardware mechanics contributing to this failure domain can be attributed to the complex interplay between the node's ASIC (Application-Specific Integrated Circuit) memory parity checks, line card operation, and power supply redundancy. The node's PE role involves high-speed packet processing and forwarding, making it susceptible to hardware faults. In this context, the ASIC memory parity errors may occur due to transient errors, overheating, or manufacturing defects. Line card crashes may be caused by software bugs, hardware malfunctions, or high-temperature conditions. Power supply redundancy loss can be triggered by a faulty power supply unit, excessive power consumption, or a power grid issue. To mitigate these risks, it is essential to promptly identify and resolve hardware/environmental faults to ensure the node's continued operation and prevent network downtime.

Pre-requisite Checks

- Verify that remote hands are dispatched to the Basavanagudi POP site.

- Check the status of redundant links to ensure they are operational.

- Confirm that the power supply units are functioning correctly and that there are no signs of overheating or other environmental issues.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to transient errors or manufacturing defects	High
Line card crashes caused by software bugs or hardware malfunctions	Medium
Power supply redundancy loss due to faulty power supply units or excessive power consumption	Low

Diagnostic Commands & Expected Outputs

```
Command: show card status, Expected Output: Card status: active, Description: This command checks the status of the line card and verifies whether it is operational.

Command: show power supply status, Expected Output: Power supply status: normal, Description: This command confirms the status of the power supply units and checks for any signs of malfunction or overheating.

Command: show error logs, Expected Output: Error log: [ASIC parity error] [timestamp], Description: This command retrieves the error logs and identifies any parity errors that may have occurred in the ASIC memory.
```

Resolution Steps

Step 1: Step: Replace faulty line card, Command: replace card slot , Description: This step involves replacing the faulty line card with a new one to restore the node's operation.

Step 2: Step: Replace faulty power supply unit, Command: replace power supply , Description: This step involves replacing the faulty power supply unit with a new one to restore power redundancy and prevent node downtime.

Step 3: Step: Perform ASIC memory parity error correction, Command: clear error logs; reset card ASIC, Description: This step involves clearing the error logs and resetting the ASIC memory to correct any parity errors and restore node operation.

Post-Check Verification

```
Command: show card status, Expected Output: Card status: active, Description: This command verifies whether the replaced line card is operational and checks the status of the node.

Command: show power supply status, Expected Output: Power supply status: normal, Description: This command confirms the status of the replaced power supply unit and checks for any signs of malfunction or overheating.

Command: show error logs, Expected Output: Error log: [none], Description: This command retrieves the error logs and verifies whether any parity errors or other issues are present.
```

Roll-back Procedures

- Step: Restore original line card, Command: replace card slot , Description: This step involves restoring the original line card to the node in case the replacement fails and the node requires the original configuration.

- Step: Restore original power supply unit, Command: replace power supply , Description: This step involves restoring the original power supply unit to the node in case the replacement fails and the node requires the original configuration.

- Step: Restore original ASIC memory parity settings, Command: reset card ASIC; clear error logs, Description: This step involves restoring the original ASIC memory parity settings and clearing the error logs in case the correction fails and the node requires the original configuration.